

Problem

Mumtaz Co is the distribution of Pakistan Tobacco Company (PTC) in Sargodha that sources its stock from PTC and distributes it to various channels downstream. In this process, a considerable manual effort goes into recording, collecting, sharing and consolidating the information on daily basis. The whole process is manually conducted from recording to information consolidation. This lack of computerization of the process results in the following problems:

1. **Delayed Information** – The manual setup requires the EoD process to close before meaningful information about the day can be available. Any unexpected event can halt this process resulting in unavailability of the information. Especially the reporting needs over the weekends or holidays.
2. **Lack of control** – the dependency on the team increases for the owner for every downstream information. This results in slowing down the decision-making process for people in-charge.
3. **Data Fragmentation and inconsistency** – Data lives in separate stations and is harder to consolidate, verify, correct and share. This necessitates a secondary process to ensure data reliability which consumes more time on daily basis. The data inconsistency issues can leak out in bookkeeping errors which trigger a long downstream process to locate and fix the smaller mistake.
4. **Lack of Reporting Structure** – Since the data collection process is manual, tedious, less structured it results in less meaningful reports. This results in the loss of meaningful and presentable information that helps in decision-making.
5. **Compliance Challenges** – Manual handling the books can make it easier to create fraudulent entries in the books causing compliance issues. The audit process can result in contractual

Moreover, I perceive that there are certain challenges that are unique to both Mumtaz Co and Tobacco Industry in Pakistan as described below.

Unique to [Tobacco] Distribution

1. **Channel policies with channel-driven pricing** – Mumtaz Co handles distribution across multiple channels including retail and wholesale which demand separate pricing for each channel. Not limited to this, filer and non-filer categories also demand separate price handling at the channel level. A serious effort is required to handle this level of complexity in distribution and bookkeeping cycle.
2. **Min. safety stock required to be maintained** – A minimal safety stock is required to be maintained by distributors all the time under the distribution contract, and to maintain that safety level the distributions must place a purchase order for the differential stock on daily

basis. This requires timely availability of true picture of the stock at hand which is both very hard and prone to inaccuracies when there is no centralized information system in place.

Unique to Mumtaz Co

1. PTC's own Mobile App and SAP portal serve as the primary source of data capturing of two most critical business events (sale and purchase) and every sale/purchase recorded on PTC system then gets copied to Mumtaz Co's local data repository. These pieces of information are timely and precisely captured and reported by the PTC systems. However, Mumtaz & Co being an independent business organization, besides being a distributor of PTC, does not have its own information system that not only records these two critical events but also streamline and support rest of the functions.

The net effect of above highlighted problems is a business running without meaningful intelligence that can affect current and future decisions that can support the organization's growth. Hence forth we present a fully integrated solution that treats information correctness and speed as its first-class citizens.

Solution Required

To address the issues highlighted above, we require an integrated software solution that automates business processes such as distribution, inventory management, purchase management, and supports performing key bookkeeping tasks including making primary accounting record of business transactions. The system should also provides a basic level of human resource management module. The interconnected nature of the solution should eliminate the data sharing overhead and reduces data corruption and inconsistency problems henceforth.

Features Required

The system should aim to deliver the smooth functioning of core business operations like distribution, channel management, purchase, bookkeeping, accounting, HR etc. Following is a detailed description of the features that are accessible to the logged in users with their credentials and privileges accordingly:

Purchase management

1. Create a purchase order aligned with the details of the order placed with PTC on its SAP portal. Each purchase created on this system will have a corresponding PTC Reference number to track.
2. Purchase return claim aligned with PTC workflow and then it should be created on this system.
3. Maintain & Control stock purchase history with PTC supply chain management system (including module).
4. View and Print purchase order using PO (internally maintained) number or PTC SAP Reference Number.
5. View and print Purchase Order history grouped by SKU, brand.

Inventory management

1. SKU Life Cycle management – Create, change, view or discontinue an SKU with the properties like pack size, stick per pack, barcode, channel wise price management.
2. Brand Life Cycle Management – create, modify the brand with specified attributes
3. Update product inventory upon purchase
4. Search the product with product code or product name.
5. Facilitation and settlement of the movement of the inventory between warehouses

6. Recording and reconciliation of the stock in transit across various channels over the day (for the purpose of sales)
7. Record and account for damaged, lost, stolen stock (ref. Claims of Mumtaz Co with PTC)
8. Cost and Price change history – will track history of cost changes and selling price changes against each SKU to facilitate the decision making around them.
9. View the inventory analytics at branch and for the business overall
10. Complete list of SKUs with search filters including SKU number, brand name
11. View SKUs master with Channel Wise pricing
12. View and Print stock at hands report filtered by SKU number, warehouse, brand

Distribution and channel management

1. Channel lifecycle management – including channel definition with properties like type (retailer, wholesale etc), Date, filer/non-filer status, price. Moreover, the lifecycle includes marking the channel active, dormant or closed. Each channel will have its own price for each SKU which can be modified by an authorized user only as and when needed.
2. Shop lifecycle management (the buyer) – register or close a retailer, or buyer, with name, address, contact details, locality, channel
3. Sales Order – Add sales order along with PTC reference number including items sold, with quantity, DR id/name, time stamp, channel, retailer details
4. Upload PTC sales data obtained from the PTC systems (Mobile App or SAP). The data upload request will be accepted by the system only if the file to be uploaded is in Excel/csv format and complies with the fixed column structure as defined in the template which will be provided along with the system.
5. Sale Return – Add, change, view, delete a sales return
6. Payment Receipts (collections) – maintain history of payments received aligned with PTC reference number for sales order
7. View and print daily (today's) sales closing summary
8. View and print sales summary grouped by salesperson, channel, (brand or SKU), warehouse
9. View and print daily distribution out and in stock summary
10. View and print daily distribution out and in stock history report
11. View and print sales order by shop/retailer/wholesale

Accounts management

Accounts module should record and report the accounting treatment of every business transaction that takes place as it is deeply integrated with all aspects of the systems on real time basis

1. Account Groups, Subgroups, and Charts of Account with unique code for each account
2. Implementation of double entry bookkeeping system to keep the business books balanced
3. Voucher system to post transactions to primary Journals. The vouchers will be limited to payment (cash or bank modes), journal voucher, expense voucher only. Distribution module related accounting entries will be embedded within the relevant features.

4. Financial statements - income, balance sheet and cash flow statement
5. Party ledgers with each transaction shown as either debit or credit respectively with a closing balance since beginning, or between two select dates
6. View and print income statement, balance sheet and cashflow statement
7. View and print party ledger
8. View and print journal vouchers
9. View and print payment vouchers

Human Resource

1. Create, view, edit and terminate an employee having fields including Full Name, CNIC, Address, role, monthly salary, commission.
2. Record monthly salaries paid to staff. The fields will be limited to month, year, salary paid, commission amount.
3. View and print monthly Salary payouts
4. View and print monthly Salary Slip based on Employee Number

Access Control

1. Fixed roles-based access control module which allows the administrator to grant or revoke the role related permissions to one or a group of users
2. Role-based access to features, individual field level access control is not included. Each role will have create/edit/view/delete/all permissions for a given feature
3. Administrator will be able to create, edit, and block users, assign or change their role

Delivery

Process

The application goes through a software development life cycle in two phases. The first one being the development phase and second being the deployment phase. In the first development phase there are multiple stages that are iterated like requirement understanding, architectural design, development and testing the system. The outcome of this process will be that the system meets the agreed upon needs for Mumtaz Co.

During the deployment process the application will be installed on your system and tested for real use cases at Mumtaz Co.

Data Transfer

The client will provide us with the closing balances of all the accounts which we will enter into the new system as opening balances. The new system will produce reports based on these closing balances and the data collected during the future operations.

Ownership and Rights

Upon full payment of the Project Fee, Mumtaz & Co. shall receive a copy of the complete source code and all business data maintained within the system. Ownership of all business data shall remain with Mumtaz & Co. Provided that no confidential information or proprietary business data of Mumtaz & Co is disclosed or reused .